Deadline: Monday, May 11, 2026, until 2pm (14:00)

Yaşar University
Spring, 2025 - 2026
SE 2232 - Software System Analysis
Asst. Prof. Dr. Deniz Özsoyeller

FINAL PROJECT

NOTES:

- **Submit your Project to Moodle "Final Project" assignment link. Do not send your project via email!**
- **Late submissions (submissions after the deadline) will not be evaluated.**
- **You can do the Project individually or as a group that includes at most 2 students.**
- **The project requires the completion of three parts: 1- Report [55 points], Implementation (Coding) [30 points], 3- Demo [15 points].**
- **Note that any copied parts from other projects will decrease your grade!**

Requirements:

- The project should be designed as a Desktop Application (with GUI) developed with Java programming language. Note that you are not designing a Web Application or a Mobile Application. The integrated development environment (IDE) that should be used is either *NetBeans* or *IntelliJ Idea*.
- For the database creation, you should use either *SQLite* or *MySQL*.

Introduction:

In this project, you will develop a software application called "**MovieCritics**" which keeps a list of the movies that the family members have watched or wish to watch individually or together.

Each *movie* should have at least the following attributes given with their data types and explanations.

- **MovieID** → INT Primary key, uniquely identifies each person in the database.
- **Title** → VARCHAR(255) (string, up to 255 characters)
- **Release year/date** → DATE (full date) or YEAR (just the year)
- **Language** → VARCHAR(100) (string), The primary spoken language(s) in the movie.
- **Country of origin** → VARCHAR(100) (string), The country where the movie was produced.
- **Genre** → VARCHAR(100) or TEXT, The category or style of the movie (e.g., drama, comedy, sci-fi).
- **DirectorId** → VARCHAR(255) or foreign key to a Person table, Stores the name of the director.
- **Watched or Not Watched** → BOOLEAN (true/false), Indicates whether a family member has watched the movie.
- **Leading Actor/Actress Id** → VARCHAR(255) or foreign key to a Person table, Stores the name.
- **Supporting Actor/Actress Id** → VARCHAR(255) or foreign key to a Person table, Stores the name.
- **About** → TEXT, A summary or description of the movie's plot or background. TEXT is used since descriptions can be long.
- **Rating** → INT (e.g., 1–10 scale), Represents the numeric score given by a user.
- **Comments** → TEXT, Stores user-written reviews or personal notes about the movie. TEXT is

appropriate since comments vary in length.

- **Poster** → VARCHAR(255) or TEXT (URL or file path)
- **ParentalRestriction** → BOOLEAN (true/false)

Each *person* should have at least the following attributes:

- **PersonID** → INT Primary key, uniquely identifies each person in the database.
- **FirstName** → VARCHAR(100) The person's given name.
- **LastName** → VARCHAR(100) The person's family name or surname.
- **DateOfBirth** → DATE The person's birthdate.
- **Nationality** → VARCHAR(100) Country of origin or citizenship.

Each *user* should have at least the following attributes:

- **UserId** (int) → Primary key, uniquely identifies each user,
- **Username** → VARCHAR(100),
- **Password** → VARCHAR(100),
- **UserType/role** → (int), e.g., mother, father, brother, sister, etc.,
- **Email** → VARCHAR(255).

Database:

In summary, you should keep at least 3 database tables: *Movie*, *Person*, and *User*. Make sure that there are at least 5 entries in each table. If you need, you can create more tables (additional to these).

LOGIN:

The application starts with **a login JFrame** (labeled as LoginFrame) where the user should enter his/her username and password to access the functionalities. There are **two types of users**. ***Type-1* - Parents/Adults:** the users who can access to functions 1-6 given below, and ***Type-2* - Children:** who can access to functions 7-11 given below. There are also the functions (i.e. Functions 12-13 given below) that both user types share.

When the user successfully logs in, the LoginFrame will disappear and instead anotherJFrame (labeled as Type1MainFrame/Type2MainFrame) will appear. MainFrame includes the main functionalities of the application that can be performed by the logged in user type. Depending on the username and password, you should determine the type of the user, and show the appropriate JFrame and the corresponding functions.

The application's functions are explained in detail below.

Parents / Adults (Admin-Level Functions)

1. **Add or Remove Movies** → Maintain the movie library by adding new entries or deleting incorrect ones.
2. **Edit Movie Details** → Update attributes like title, about, or director.
3. **Manage User Accounts** → Create accounts for family members, assign roles, and reset passwords.
4. **Set Viewing Rules** → Mark age restrictions.
5. **Moderate Content** → Approve or remove inappropriate comments or ratings.

6. **View Family Analytics** → See statistics such as most-watched movies and average ratings.

Children (Member-Level Functions)

7. **Log Watched Movies** → Mark movies as "watched" or "not watched."
8. **Rate Movies** → Give star ratings or scores (e.g., 1–5 stars).
9. **View and Add Comments** → Write short reviews.
10. **Create Personal Watchlists** → Save movies they want to watch later.
11. **Track Progress** → See how many movies they have watched compared to siblings.

Shared Functions (Both Roles)

12. **Browse Movies** → Search and filter by genre, director, or release year, and display all movie details. Note that Chidren can only view the details of the movie where the value of ParentalRestriction attribute is False.
13. **View Family Ratings** → See how other family members rated a movie.

NOTE: How you design the GUI (Graphical User Interface) of your project and the appearance of the common functions shared by both user types are optional. Make sure that it is easy to use and the components on the frames are organized well and understandable.

Report:

- You should use the project template in Moodle to prepare your project report. The report consists of eight sections to be completed.
- Note that you should use Visual Paradigm to draw the UML Diagrams required in the project. Put your diagram as an image in your report.

Presentation:

- Prepare a demonstration (demo) video of your project in English.
- The video should only show the application (the codes, database tables and GUIs). You do not need to explain any section from your Report.
- Briefly introduce yourself and the goal of the project.
- Briefly explain each of your class in your implementation and its purpose.
- Test each function/feature of your application (including your database) and explain in which class the tested function is implemented in your project.
- **Duration:** 5-10 minutes.
- **Format:** mp4
- If the project is completed as a group, then each group member should present, and the presentation time should be divided equally amongst the group members.
- If the project is completed as a group, the submission of one group member is enough.
- Check to see if your demo video is working or not before submitting it. Videos that cannot be opened, do not function properly, or are in formats other than MP4 will not be evaluated.

Project Submission:

- Submit the parts below in a **SINGLE FOLDER** compressed in zip/rar format or directly upload them to Moodle one by one without putting them in a compressed folder to Moodle Assignment link "SE 2232 Final Project".

1. **Report file (Format: .pdf/.doc/.docx)**
2. **Demo file (Format: .mp4)**
3. **Java codes (Format: ONLY the files with .java or .form extension)**
4. **Visual Pradigm files (Format: .vpp)**
5. **Database files**

- **IMPORTANT NOTE:** DO NOT CREATE SUBFOLDERS OR SUBMIT YOUR FILES WITHIN SUBFOLDERS!